Simulating a Particle Detector and Particle Decay in C++

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Abstract

The aim of this project was to utilize object oriented programming practices to create a particle detector and particle decay simulators in C++. It consists of two main inheritance chains, one for the particle classes and another for sub-detector classes. These classes, and their respective container classes ParticleBunch and Detector, practice encapsulation by storing data members as private or protected with access via public member functions. The use of virtual and pure virtual functions in the abstract parent classes of the hierarchy enables polymorphism in the child classes.

1 Introduction

Object oriented programming (OOP) is built on four pillars: encapsulation, inheritance, polymorphism and abstraction.

2 Theory

3 Results

```
int main()
{

        return 0;
}
```